

Chapter 22

Waste Management — II Safe Disposal of Waste

Class 10 - Total Geography Morning Star

Choose the correct option

Question 1

Dry waste consists of waste that is

1. biodegradable
2. non-biodegradable
3. non-toxic
4. organic

Answer

non-biodegradable

Question 2

Wet waste is the waste that is in nature.

1. non-biodegradable
2. organic
3. recyclable
4. toxic

Answer

organic

Question 3

..... of waste is the key to effective waste management.

1. Segregation

2. Modification
3. Solidification
4. Nitrification

Answer

Segregation

Question 4

Biodegradable waste can be converted into

1. soil
2. humus
3. rubber
4. compost

Answer

compost

Question 5

..... is the cheapest method of waste disposal.

1. Open dumping
2. Sanitary landfill
3. Incineration
4. Recycling

Answer

Open dumping

Question 6

..... is a way of disposing waste on land without causing health hazards or public safety.

1. Dumping
2. Sanitary landfill
3. Recycling
4. Incineration

Answer

Sanitary landfill

Question 7

..... is an aerobic method of decomposing solid wastes.

1. Incineration
2. Recycling
3. Composting
4. Dumping

Answer

Composting

Question 8

The organic waste from households undergo to form compost.

1. reduction
2. burning
3. decomposition
4. churning

Answer

decomposition

Question 9

Compositing rejuvenates poor soils by adding

1. nutrients
2. manure
3. humus
4. oxygen

Answer

humus

Question 10

..... process is used for waste disposal by industries and municipalities.

1. Compositing
2. Dumping
3. Incineration
4. Segregation

Answer

Incineration

Question 11

Waste can be effectively managed by using three of waste management.

1. S's
2. R's
3. C's
4. M's

Answer

R's

Question 12

Which of the following is Not included in three R's of waste management?

1. Reduce
2. Reuse
3. Recycle
4. Regenerate

Answer

Regenerate

Answer the following questions

Question 1(a)

What do you mean by segregation of waste? How does it ensure safe waste disposal?

Answer

Segregation of waste means dividing the waste into different categories like dry and wet or biodegradable and non-biodegradable.

When waste is segregated, there occurs a reduction in the volume of waste that reaches the landfills. Air and water pollution is considerably reduced and it becomes easier to apply different processes of waste disposal like composting, recycling and incineration.

Question 1(b)

What is meant by open dumping of waste? Why is open dumping not considered as an environment friendly method?

Answer

In open dumping, waste materials are dumped in open low lands far away from the city.

This method is not environment friendly as the open pits spoil the sight of the area and become a breeding ground for mosquitoes, flies, insects, etc., that are the carriers of harmful diseases. They give out foul odour. The burning of waste material in the open dumps pollutes the air.

Another danger of open dumping is that rainwater could carry the harmful substances to the nearby streams, ponds or lakes and if the water seeps down it could pollute the groundwater.

Question 1(c)

Give a reason for each of the following:

- (i) Vehicles used for transporting waste should be covered.
- (ii) Waste segregation is key to waste management.
- (iii) Roads from plastic waste are environment friendly.

Answer

(i) Vehicles used for transporting waste should be covered as this prevents the wastes from being scattered. Waste should not be visible to public, nor exposed to open environment.

(ii) Waste segregation is key to waste management as when waste is segregated, there occurs a reduction in the volume of waste that reaches the landfills. Air and water pollution is considerable reduced and it becomes easier to apply different processes of waste disposal like composting, recycling and incineration.

(iii) Roads from plastic waste are environment friendly because of the following reasons-

1. It overcomes the growing problem of plastic waste disposal in the country. Since a large amount of plastic waste is required for a small stretch of road, the amount of waste plastic strewn around gets tremendously reduced.

2. This road construction process is eco-friendly with no toxic gases being released.

Question 1(d)

- (i) What is a sanitary landfill?
- (ii) How is waste disposed of in a sanitary landfill?

Answer

(i) Sanitary landfill is a way of disposing refuse on land without creating nuisances or hazards to public health and safety. The waste disposal is carried out with minimal environmental damage and in areas already spoiled or in need of restoration.

(ii) In sanitary landfill, the waste undergoes the following five phases for safe disposal-

1. In the first phase of operation, aerobic bacteria depletes the available oxygen and causes the temperature to rise.
2. In the second phase, anaerobic conditions get established and lead to the evolution of hydrogen and carbon dioxide.
3. Phase three establishes population of bacteria and the beginning of methanogenic activity, i.e., production of methane from the decomposition of organic matter.
4. In the fourth phase, the methanogenic activity becomes stabilised.
5. The fifth phase depletes the organic matter, and the system returns to aerobic state.

Question 2(a)

What is biodegradable waste? Name two useful products which can be made from biodegradable domestic waste.

Answer

Waste which is organic in nature and can be decomposed or broken down by living organisms is called biodegradable waste. For example- stale food, rotten fruits and vegetables etc.

Two useful products which can be made from biodegradable domestic waste are compost and biogas.

Question 2(b)

What service is indirectly done by the rag-pickers for the disposal of waste?

Answer

The poor rag pickers make a living by sorting out reusable and recyclable items from discarded solids. They do a good job by removing much of the waste from the garbage dumps. Pieces of

metal, glass, rubber, plastics etc., are removed to be recycled to get finished products. Thus, they segregate the waste and make waste disposal easier.

Question 2(c)

State three steps that should be taken by the municipal authorities for the safe disposal of solid waste.

Answer

The following three steps should be taken by the municipal authorities for the safe disposal of solid waste-

1. Organising house to house collection of municipal solid wastes.
2. Biomedical waste and industrial waste should not be mixed with municipal solid waste. The waste should be segregated into biodegradable and non-biodegradable.
3. Stray animals should not be allowed to move around waste disposal facilities.

Question 2(d)

What is composting? Give two advantages of using compost.

Answer

Composting of waste in an aerobic method of decomposing solid wastes. The organic wastes from households are made to undergo decomposition in such a way that bacteria and other micro-organisms break them down and produce a safe, clean and soil like material called compost.

Two advantages of using compost are-

1. It enhances soil nutrients and water retention capacity of soil.
2. It suppresses plant diseases.

Question 3(a)

What are the three R's of waste management?

Answer

The three R's of waste management are-

1. Reducing the waste
2. Reusing the waste
3. Recycling the waste

Question 3(b)

Explain how reusing waste can reduce the burden of waste disposal.

Answer

One way to prevent waste generation, improve our communities, and increase the material well-being of our citizens is to take useful products discarded by those who no longer want or need them and provide them to those who do.

Reusing an item means that it continues to be a valuable, useful, productive item, and replaces new items that would utilize more water, energy, timber, petroleum, and other limited natural resources in their manufacture. For example, instead of discarding old or torn shoes, they can be given to under-privileged or poor people for use.

Hence, if we reuse products, we ultimately avoid the generation of waste and reduce the burden of waste disposal.

Question 3(c)

Give a reason for each of the following:

- (i) Recycling of waste to produce paper can reduce deforestation.
- (ii) Sugarcane waste can be recycled into useful products.
- (iii) We should avoid using polythene carry bags.

Answer

(i) Recycling of waste to produce paper can reduce deforestation as in India, we have tonnes of bagasse from sugarcane during a particular season. Bagasse is used in the manufacture of paper pulp and this helps to save trees which are normally used for making paper pulp.

(ii) Bagasse is a waste product from the sugar industry. In India, we have tonnes of bagasse from sugarcane during a particular season. Bagasse is used in the manufacture of paper pulp and this helps to save trees which are normally used for making paper pulp. Bagasse is also used for making packaging material for dairy products. Hence, sugarcane waste can be recycled into useful products.

(iii) We should avoid using polythene carry bags because plastic is non-biodegradable. Plastic has to be incinerated, recycled or buried in landfills. The polythene carry bags are made from recycled plastic. They are harmful because the melting of plastic and plastic products breaks some polymer chains into smaller units which are harmful.

Question 3(d)

What is meant by recycling of waste effectively? Give one example.

Answer

Recycling is a process by which wastes are converted into reusable products, thereby reducing the usage of raw material and energy and controlling air, water and soil pollution.

Recycling of waste effectively means to recycle an item till it cannot be recycled further and cannot be utilised any more.

For example, in India, we have tonnes of bagasse from sugarcane during a particular season. Bagasse is used in the manufacture of paper pulp and this helps to save trees which are normally used for making paper pulp.

Thinking Skills

Question 1

How can segregation of waste into biodegradable and non-biodegradable reduce the quantity of waste and can be utilised for generating sustainable energy?

Answer

Segregation allows for targeted management of waste. By separating biodegradable and non-biodegradable waste, the overall quantity of waste sent to landfills decreases, reducing environmental impact. The segregated waste can be managed as follows:

1. Composting — Biodegradable waste can be composted, converting it into nutrient-rich compost. Composting diverts organic waste from landfills, improves soil health, and reduces the need for chemical fertilizers.
2. Waste-to-energy conversion — Non-biodegradable waste, like certain plastics, can be used for waste-to-energy processes such as incineration or gasification. This generates sustainable energy, reducing reliance on fossil fuels.
3. Anaerobic decomposition — Biodegradable waste, especially organic waste, can undergo anaerobic decomposition to produce biogas, a renewable energy source for heating, cooking, or electricity generation.
4. Efficient waste management — Segregation streamlines reusing, recycling and disposal processes, improving overall waste management efficiency and promoting effective recycling and recovery of materials from non-biodegradable waste.

Question 2

Bio-fuels should be developed on a large scale as an alternative to fossil fuels and to reduce generation of waste. State your views.

Answer

Developing biofuels on a large scale as an alternative to fossil fuels and to reduce waste generation can be beneficial for several reasons:

1. **Climate Change Mitigation** — Biofuels have lower carbon emissions, reducing greenhouse gas impact. Their production and combustion release CO₂, but the CO₂ is offset by the absorption of CO₂ during biomass growth.
2. **Waste Reduction** — Utilizing organic waste for biofuel production helps reduce waste accumulation and promotes a circular economy by converting waste into a valuable resource.
3. **Energy Independence** — Biofuels decrease reliance on fossil fuel imports, enhancing energy security. It provides an alternative and domestically sourced fuel option.
4. **Rural Development and Job Creation** — Biofuel production supports agricultural activities and creates employment opportunities in farming, biomass supply chains, and biofuel production facilities.
5. **Technological Advancements** — Investment in biofuel research drives innovation for more efficient and sustainable production processes.

Question 3

Find out recent innovations by which waste has been used to form useful products.

Answer

Recent innovations by which waste has been used to form useful products are as follows:

1. **Waste-to-Energy** — Advanced technologies convert waste into electricity, heat, or biofuels, reducing waste accumulation and generating renewable energy.
2. **Upcycling** — Innovative approaches transform waste materials into valuable products, such as building materials or clothing, through recycling and upcycling techniques.
3. **Circular Economy** — Companies adopt circular economy models, where waste from one process becomes a resource for another, such as converting food waste into compost for agricultural use.
4. **Bioplastics** — Researchers develop bioplastics from organic waste, offering biodegradable alternatives to conventional plastics.
5. **Carbon Capture and Utilization** — Innovations capture carbon emissions and convert them into useful products, like using CO₂ in the production of construction materials.

Question 4

What are eco bricks? Make a list of items that you can make using eco bricks.

Answer

Eco bricks or ecological bricks or plastic bricks, are building blocks made by stuffing plastic waste tightly into plastic bottles. These bricks provide an eco-friendly solution to manage plastic waste.

Eco bricks can be used to make the following items:

1. Furniture — They can be used in making chairs, benches, tables, and shelves.
2. Garden and landscaping structures — They can be used to build garden walls, raised garden beds, planters, and decorative structures like trellises and sculptures.
3. Pavements and pathways — Eco bricks can be used to construct durable and sustainable pavements and pathways in outdoor areas.
4. Partition walls — They can be used to build interior partition walls in homes or other buildings.
5. Playgrounds and recreational structures — Eco bricks can be used to create play structures, such as slides, climbing walls, and seating areas, in parks and playgrounds.
6. Art installations — Eco bricks can be incorporated into artistic installations and sculptures, showcasing creativity while promoting environmental awareness.